

Recursive Factorizations

The Category of Ranked Recursive Factorization Systems

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Abstract

We develop a categorical theory of *ranked recursive factorization systems* (RRFS). The motivating examples are numerical Pratt trees, the prime-index polynomial family f_n , the polynomial Pratt forests of arbitrary monic rational polynomials, finite truncations over finite fields, finite directed acyclic multigraphs, hierarchical bills of materials, and deterministic software dependency architectures. An RRFS is a free commutative factorization monoid together with a predecessor assignment on atoms and a rank strictly decreasing along every predecessor factor. The predecessor data determine finite recursive trees, a transition operator B , an incidence resolvent $C = (I - B)^{-1}$, total recursive coordinates, a boundary map $\beta = I - B$, and a telescoping reconstruction formula in the group completion.

After a large gallery of finite, infinite, connected, decomposable, coupled, arithmetic, polynomial, and engineered examples, we define strict and linear categories of RRFS. We prove functoriality of transition, incidence, boundary, coordinate, forest, metric, topological, and front-polynomial constructions. Root coordinates and total coordinates form naturally isomorphic functors, with β and C as inverse natural transformations. Inspired by the standard language of topology, groups, rings, and modules, we introduce recursive subsystems, hereditary and co-hereditary subsystems, quotient systems, kernel congruences, normal subsystems in the linear sense, direct sums, products, extensions, composition series, indecomposable systems, and several notions of simplicity. We identify finite RRFS with finite acyclic directed multigraphs carrying multiplicities and show how coupled block systems behave. The map $p \mapsto f_p$ is the prototypical isomorphism between two apparently different RRFS, and evaluation at $x = 2$ gives its inverse on the prime-index subsystem. The final sections explain how the formalism applies to software versioning and dependency provenance and formulate structural classification problems.

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1 A gallery of recursive factorization systems

The theory is easiest to understand through examples. In every example there is a collection of atomic types, each atom has a finite multiset of predecessor atoms, and repeated predecessor expansion terminates because a rank strictly decreases.

1.1 Positive integers and numerical Pratt trees

Let the atoms be the primes $\mathbb{P}$. Define

$$D(2) = 1, \quad D(p) = p - 1 \quad (p > 2).$$

If

$$p - 1 = \prod_{q \in \mathbb{P}} q^{v_q(p-1)},$$

then the tree T_p has root label p and $v_q(p - 1)$ child copies of T_q . For

$$n = \prod_p p^{v_p(n)},$$

the forest F_n is the disjoint union of $v_p(n)$ copies of T_p . The total number of q -vertices is denoted $m_q(n)$ and satisfies

$$\begin{aligned} m_q(1) &= 0, & m_q(ab) &= m_q(a) + m_q(b), \\ m_q(p) &= \mathbf{1}_{q=p} + m_q(p - 1). \end{aligned}$$

The numerical telescoping formula is

$$n = \prod_p \left(\frac{p}{p-1} \right)^{m_p(n)} = \prod_p \left(1 - \frac{1}{p} \right)^{-m_p(n)}. \quad (1)$$

The terminal atom is 2, and the terminal coordinate satisfies the sharp bounds

$$\log_3 n \leq m_2(n) \leq \log_2 n. \quad (2)$$

1.2 The prime-index polynomial system

Define

$$f_1(x) = 1, \quad f_2(x) = x,$$

and for odd primes p ,

$$f_p(x) = 1 + \prod_{q|p-1} f_q(x)^{v_q(p-1)}. \quad (3)$$

Extend multiplicatively:

$$f_n(x) = \prod_p f_p(x)^{v_p(n)}.$$

Then

$$f_{mn} = f_m f_n, \quad f_n(2) = n,$$

and, with the polynomial predecessor defined below,

$$D(f_p) = f_{p-1}.$$

The prime-index polynomials are pairwise distinct irreducibles. The assignment

$$p \mapsto f_p$$

therefore transports the numerical Pratt recursion exactly into a recursively closed irreducible polynomial subsystem.

1.3 General polynomial Pratt forests over $\mathbb{Q}$

Let

$$\mathcal{I}_{\mathbb{Q}} = \{\varphi \in \mathbb{Q}[x] : \varphi \text{ monic, nonconstant, irreducible}\}.$$

Define

$$D\varphi = \begin{cases} 1, & \varphi = x, \\ \varphi - \varphi(0), & \varphi \neq x. \end{cases} \quad (4)$$

For $\varphi \neq x$, the polynomial $D\varphi$ is monic and divisible by x . Every irreducible factor of $D\varphi$ has strictly smaller degree than φ . Hence degree is a rank.

If

$$D\varphi = \prod_{\psi \in \mathcal{I}_{\mathbb{Q}}} \psi^{B_{\psi, \varphi}},$$

then T_{φ} has $B_{\psi, \varphi}$ children of type ψ . For a monic polynomial

$$f = \prod_{\varphi} \varphi^{v_{\varphi}(f)},$$

define $M_{\psi}(f)$ as the total number of ψ -vertices in its forest. Then

$$M_{\psi}(fg) = M_{\psi}(f) + M_{\psi}(g)$$

and

$$f = \prod_{\varphi \in \mathcal{I}_{\mathbb{Q}}} \left(\frac{\varphi}{D\varphi} \right)^{M_{\varphi}(f)}. \quad (5)$$

Although the local factors are rational functions, the recursively consistent product telescopes to f .

1.4 Finite-field polynomial systems

Fix a finite field $\mathbb{F}_q$ and let $\mathcal{I}_{q, \leq d}$ be the finite set of monic irreducibles in $\mathbb{F}_q[x]$ of degree at most d . Use the same predecessor convention

$$D\varphi = \varphi - \varphi(0), \quad Dx = 1.$$

Every predecessor factor has smaller degree, so $\mathcal{I}_{q, \leq d}$ is descendant-closed. Thus it defines a finite RRFS. Its number of atomic types is

$$\#\mathcal{I}_{q, \leq d} = \sum_{k=1}^d \frac{1}{k} \sum_{r|k} \mu(r) q^{k/r}. \quad (6)$$

The telescoping reconstruction remains valid in $\mathbb{F}_q(x)^{\times}$. Differential statements require care in characteristic p , but the recursive factorization theory itself does not.

1.5 A finite chain

Let

$$\mathcal{A} = \{\tau, a_1, \dots, a_h\}, \quad D\tau = 1, \\ Da_1 = \tau^{r_1}, \quad Da_j = a_{j-1}^{r_j} \quad (2 \leq j \leq h),$$

where $r_j \geq 1$. Then

$$M_{a_j}(a_h) = \prod_{k=j+1}^h r_k, \quad M_\tau(a_h) = \prod_{k=1}^h r_k.$$

This is a finite multilevel assembly chain.

1.6 A finite branching system

Let

$$D\tau = 1, \quad Da = \tau^2, \quad Db = \tau^3, \quad Dc = a^2b.$$

In the order (τ, a, b, c) ,

$$B = \begin{pmatrix} 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B^3 = 0.$$

The c -tree has node vector

$$M(c) = (7, 2, 1, 1)^T.$$

This example already contains nontrivial branching, multiplicity, boundary cancellation, and a finite front polynomial.

1.7 Finite acyclic multigraphs

Let $G = (V, E)$ be a finite directed acyclic multigraph. For each $v \in V$, introduce an atom a_v and define

$$D(a_v) = \prod_{u \in V} a_u^{m(u,v)}, \tag{7}$$

where $m(u, v)$ is the number of directed edges $v \rightarrow u$. A topological ordering gives a rank. Conversely, every finite RRFS determines such a graph. Finite RRFS are therefore algebraic presentations of finite acyclic directed multigraphs with multiplicity.

1.8 Disjoint unions and decomposable systems

If $\mathcal{R}_1$ and $\mathcal{R}_2$ have disjoint atom sets, their disjoint union has block-diagonal transition matrix

$$B_{1 \sqcup 2} = \begin{pmatrix} B_1 & 0 \\ 0 & B_2 \end{pmatrix}.$$

Its forests are independent disjoint unions. This is the basic decomposable example.

1.9 One-way coupled systems

Let

$$B = \begin{pmatrix} B_1 & K \\ 0 & B_2 \end{pmatrix}$$

with nonnegative integer entries and with an ordering making B strictly triangular. Then

$$(I - B)^{-1} = \begin{pmatrix} C_1 & C_1 K C_2 \\ 0 & C_2 \end{pmatrix}, \quad C_i = (I - B_i)^{-1}. \quad (8)$$

The off-diagonal block $C_1 K C_2$ counts paths that descend in system 2, cross the coupling once, and continue in system 1.

1.10 Software dependency architectures

Fix package-version types P . Let $b_{Q,P}$ be the multiplicity of direct dependency Q inside package P , and define

$$D(P) = \prod_Q Q^{b_{Q,P}}.$$

After cycle condensation, a deterministic dependency architecture is a finite DAG and hence a finite RRFS. If u is the root-package vector and x is the unfolded total occurrence vector, then

$$x = (I - B)^{-1}u, \quad u = (I - B)x. \quad (9)$$

This separates directly declared roots from transitively forced dependencies.

1.11 Bills of materials, attack trees, and grammars

The same construction applies to hierarchical bills of materials, finite fault trees, attack trees, acyclic context-free production systems, and typed workflow decompositions. In each case the free commutative monoid records multiplicity of root objects; the predecessor map records direct children; and recursive expansion produces an occurrence forest.

2 Ranked recursive factorization systems

Definition 2.1 (Ranked recursive factorization system). A *ranked recursive factorization system* is a tuple

$$\mathcal{R} = (\mathcal{A}, \mathcal{M}, D, \rho),$$

where:

- (i) $\mathcal{A}$ is a set of atoms;
- (ii) $\mathcal{M} = \mathbb{N}_0^{(\mathcal{A})}$ is the free commutative monoid on $\mathcal{A}$, written multiplicatively;
- (iii) $D : \mathcal{A} \rightarrow \mathcal{M}$ assigns a finite predecessor factorization to every atom;

(iv) $\rho : \mathcal{A} \rightarrow \mathbb{N}_0$ is a rank such that

$$\beta \in \text{supp}(D\alpha) \implies \rho(\beta) < \rho(\alpha).$$

An atom τ is terminal if $D\tau = 1$.

Write

$$D\alpha = \prod_{\beta \in \mathcal{A}} \beta^{B_{\beta,\alpha}}. \quad (10)$$

The nonnegative integer matrix B is the transition matrix.

Proposition 2.2 (Local nilpotence). *For every finitely supported vector $x \in \mathbb{Z}^{(\mathcal{A})}$, one has $B^k x = 0$ for all sufficiently large k .*

Proof. The support of x has bounded rank. Every application of B strictly lowers rank along each nonzero contribution. After more steps than the maximal rank appearing in the support, no contribution remains. $\square$

Definition 2.3 (Atomic trees and object forests). The tree T_α has root α and, for each β , exactly $B_{\beta,\alpha}$ child copies of T_β . For

$$a = \prod_{\alpha} \alpha^{v_\alpha(a)},$$

the forest F_a is the disjoint union of $v_\alpha(a)$ copies of T_α .

Definition 2.4 (Root and total coordinates). The root vector is

$$v(a) = (v_\alpha(a))_{\alpha \in \mathcal{A}}.$$

The total recursive coordinate vector is

$$M(a) = (M_\alpha(a))_{\alpha \in \mathcal{A}},$$

where $M_\alpha(a)$ counts all α -labelled nodes in F_a .

Theorem 2.5 (Boundary and incidence identities). *For every $a \in \mathcal{M}$,*

$$M(a) = v(a) + BM(a), \quad (11)$$

so

$$\boxed{v(a) = (I - B)M(a), \quad M(a) = (I - B)^{-1}v(a).} \quad (12)$$

Moreover,

$$C := (I - B)^{-1} = I + B + B^2 + \dots \quad (13)$$

terminates on every finitely supported vector.

Proof. Every β -node is either a root, counted by $v_\beta(a)$, or a child of some α -node. There are $B_{\beta,\alpha}$ such children for each α -node, giving

$$M_\beta(a) = v_\beta(a) + \sum_{\alpha} B_{\beta,\alpha} M_\alpha(a).$$

Local nilpotence gives the finite geometric inverse. $\square$

Corollary 2.6 (Path expansion). *For atoms β, α ,*

$$C_{\beta, \alpha} = \sum_{k \geq 0} (B^k)_{\beta, \alpha} = \sum_{\gamma: \alpha \rightsquigarrow \beta} \prod_{e \in \gamma} m(e), \quad (14)$$

where the sum runs over all directed descending paths and $m(e)$ denotes edge multiplicity.

Definition 2.7 (Recursive cone). The natural coordinate cone is

$$\mathcal{C}_{\mathcal{R}} = M(\mathcal{M}) \subseteq \mathbb{N}_0^{(\mathcal{A})}.$$

Theorem 2.8 (Intrinsic cone characterization). *For finitely supported $x \in \mathbb{Z}^{(\mathcal{A})}$, the following are equivalent:*

- (i) $x = M(a)$ for some $a \in \mathcal{M}$;
- (ii) $x \geq 0$ and $(I - B)x \geq 0$;
- (iii) $x = Cu$ for some $u \in \mathbb{N}_0^{(\mathcal{A})}$.

2.1 Group completion and telescoping reconstruction

Let

$$G(\mathcal{M}) = \mathbb{Z}^{(\mathcal{A})}$$

be the Grothendieck group of the free commutative monoid. Identify an atom with its basis vector in this group and define the local increment

$$\Delta(\alpha) = \alpha D(\alpha)^{-1}.$$

Theorem 2.9 (Abstract telescoping reconstruction). *For every $a \in \mathcal{M}$,*

$$\boxed{a = \prod_{\alpha \in \mathcal{A}} \Delta(\alpha)^{M_{\alpha}(a)}}. \quad (15)$$

Proof. The exponent of a fixed atom β on the right is

$$M_{\beta}(a) - \sum_{\alpha} B_{\beta, \alpha} M_{\alpha}(a) = ((I - B)M(a))_{\beta} = v_{\beta}(a).$$

Thus both sides have identical atomic exponent vectors. □

2.2 Hilbert geometry

Define

$$\Phi_{\mathcal{R}}(a) = M(a) \in \ell^2(\mathcal{A}).$$

Every vector is finitely supported.

Proposition 2.10. *The map $\Phi_{\mathcal{R}}$ is injective and satisfies*

$$\Phi_{\mathcal{R}}(ab) = \Phi_{\mathcal{R}}(a) + \Phi_{\mathcal{R}}(b).$$

Hence

$$K_{\mathcal{R}}(a, b) = \sum_{\alpha} M_{\alpha}(a) M_{\alpha}(b)$$

is positive semidefinite, and

$$d_{\mathcal{R}}(a, b) = \|M(a) - M(b)\|_2$$

is a metric.

Corollary 2.11 (Translation invariance). *For all $a, b, c \in \mathcal{M}$,*

$$d_{\mathcal{R}}(ac, bc) = d_{\mathcal{R}}(a, b).$$

3 Finite systems and their graph classification

Theorem 3.1 (Finite classification). *For a finite atom set, the following data are equivalent:*

- (i) *a ranked recursive factorization system;*
- (ii) *a nilpotent matrix $B \in M_N(\mathbb{N}_0)$;*
- (iii) *a finite acyclic directed multigraph with multiplicities.*

Proof. A rank strictly decreases along every edge, so cycles are impossible. Conversely, every finite DAG has a topological ordering. The corresponding multiplicity matrix is strictly triangular and therefore nilpotent. Its edge relation defines the predecessor monomials. $\square$

Remark 3.2. The word “factorization” refers to the free commutative monoid presentation, not necessarily to multiplication in a pre-existing ring. A ring realization is additional structure.

4 The strict category RRFS

Definition 4.1 (Strict morphism). Let $\mathcal{R} = (\mathcal{A}, \mathcal{M}, D, \rho)$ and $\mathcal{S} = (\mathcal{A}', \mathcal{M}', D', \rho')$. A strict morphism

$$F : \mathcal{R} \rightarrow \mathcal{S}$$

is a map $f : \mathcal{A} \rightarrow \mathcal{A}'$ whose multiplicative extension $\widehat{f} : \mathcal{M} \rightarrow \mathcal{M}'$ satisfies

$$\widehat{f}(D\alpha) = D'(f(\alpha)) \quad (\alpha \in \mathcal{A}). \quad (16)$$

Optionally one may require $\rho'(f(\alpha)) \leq \rho(\alpha)$; rank preservation is not required for the algebraic identities.

The induced pushforward on coordinate vectors is

$$(f_*x)_{\gamma} = \sum_{\alpha: f(\alpha)=\gamma} x_{\alpha}. \quad (17)$$

Theorem 4.2 (Intertwining identities). *A strict morphism satisfies*

$$f_* B_{\mathcal{R}} = B_{\mathcal{S}} f_*, \quad (18)$$

$$f_*(I - B_{\mathcal{R}}) = (I - B_{\mathcal{S}}) f_*, \quad (19)$$

and

$$f_* C_{\mathcal{R}} = C_{\mathcal{S}} f_*. \quad (20)$$

Consequently,

$$f_* M_{\mathcal{R}}(a) = M_{\mathcal{S}}(\widehat{f}(a)). \quad (21)$$

Definition 4.3. The category whose objects are RRFS and whose arrows are strict morphisms is denoted **RRFS**.

Identities and compositions are inherited from maps of atom sets, and predecessor compatibility is stable under composition.

4.1 Monomorphisms, epimorphisms, and isomorphisms

Proposition 4.4. *A strict morphism injective on atoms is a monomorphism in **RRFS**. A bijective strict morphism is an isomorphism. If f is surjective on atoms, then it is an epimorphism among strict morphisms whose underlying atom maps are ordinary set maps.*

Remark 4.5. Categorical epimorphisms can be subtler in enriched variants. The stated criterion is sufficient and is exact in the concrete strict category considered here.

5 The linear category **LinRRFS**

Strict atom maps are sometimes too rigid. A package type may map to several abstract component types, or several detailed types may be aggregated.

Definition 5.1 (Linear recursive morphism). A linear morphism $H : \mathcal{R} \rightarrow \mathcal{S}$ is a nonnegative integer matrix

$$H : \mathbb{Z}^{(\mathcal{A})} \rightarrow \mathbb{Z}^{(\mathcal{A}')}$$

with finite columns satisfying

$$\boxed{HB_{\mathcal{R}} = B_{\mathcal{S}}H.} \quad (22)$$

Proposition 5.2. *Linear recursive morphisms compose, contain identities, and form a category **LinRRFS**. Every strict morphism determines a linear morphism by its pushforward matrix.*

Theorem 5.3 (Linear naturality). *For every linear recursive morphism,*

$$H\beta_{\mathcal{R}} = \beta_{\mathcal{S}}H, \quad HC_{\mathcal{R}} = C_{\mathcal{S}}H. \quad (23)$$

In particular, valid total-coordinate vectors are transported to valid total-coordinate vectors whenever H also preserves the nonnegative root cone.

6 Functorial structures and natural transformations

6.1 Underlying graph, monoid, and coordinate functors

There are natural functors:

$$\text{Graph} : \mathbf{RRFS} \rightarrow \mathbf{DAGMulti},$$

$$\text{Root, Total} : \mathbf{RRFS} \rightarrow \mathbf{CMon},$$

where $\text{Root}(\mathcal{R}) = \mathbb{N}_0^{(A)}$ uses root coordinates and $\text{Total}(\mathcal{R}) = \mathcal{C}_{\mathcal{R}}$ uses total recursive coordinates.

The boundary and incidence operators define component maps

$$\beta_{\mathcal{R}} = I - B_{\mathcal{R}} : \text{Total}(\mathcal{R}) \rightarrow \text{Root}(\mathcal{R}),$$

$$C_{\mathcal{R}} = (I - B_{\mathcal{R}})^{-1} : \text{Root}(\mathcal{R}) \rightarrow \text{Total}(\mathcal{R}).$$

Theorem 6.1 (Natural coordinate equivalence). *The families $\beta = (\beta_{\mathcal{R}})_{\mathcal{R}}$ and $C = (C_{\mathcal{R}})_{\mathcal{R}}$ are inverse natural transformations*

$$\text{Total} \xRightarrow{\beta} \text{Root}, \quad \text{Root} \xRightarrow{C} \text{Total}.$$

Thus root coordinates and total recursive coordinates are naturally isomorphic functors.

6.2 The forest functor

Let **RootForest** denote the category of finite typed rooted forests and label-respecting forest maps induced by strict RRFS morphisms.

Proposition 6.2. *The assignment*

$$(\mathcal{R}, a) \longmapsto F_a$$

is functorial: strict morphisms relabel every node and preserve parent-child multiplicities after identified labels are aggregated.

6.3 The Hilbert feature functor

For finite systems, every linear morphism H induces a bounded linear map between finite-dimensional Hilbert spaces. For infinite systems, boundedness follows under standard uniform column-sum or operator-norm hypotheses.

Proposition 6.3. *If H is an isometric coordinate embedding, then*

$$K_{\mathcal{R}}(a, b) = K_{\mathcal{S}}(Ha, Hb), \quad d_{\mathcal{R}}(a, b) = d_{\mathcal{S}}(Ha, Hb).$$

More generally, bounded H gives

$$d_{\mathcal{S}}(Ha, Hb) \leq \|H\| d_{\mathcal{R}}(a, b).$$

7 Topological language

The analogy with topological spaces becomes literal through the descendant order.

Definition 7.1 (Descendant preorder). For atoms β, α , write

$$\beta \preceq \alpha$$

if β occurs in T_α , equivalently if $C_{\beta, \alpha} > 0$.

Because rank decreases along proper edges, this is a partial order.

Definition 7.2 (Alexandrov topology). Equip $\mathcal{A}$ with the Alexandrov topology whose open sets are upward-closed sets for $\preceq$, or dually whose closed sets are descendant-closed sets.

Proposition 7.3. *Every strict morphism is order-preserving and therefore continuous for the Alexandrov topology.*

Proof. If $\beta \preceq \alpha$, a descending path from α to β is mapped to a descending path, possibly with identified labels, from $f(\alpha)$ to $f(\beta)$. Thus $f(\beta) \preceq f(\alpha)$. $\square$

Remark 7.4. A descendant-closed subsystem is exactly a closed subspace in this topology. This gives a precise topological interpretation of recursive subsystems.

8 Subsystems and inclusions

Definition 8.1 (Recursive subsystem). A subset $\mathcal{A}_0 \subseteq \mathcal{A}$ is descendant-closed if

$$\alpha \in \mathcal{A}_0, \beta \in \text{supp}(D\alpha) \implies \beta \in \mathcal{A}_0.$$

The restricted data define a recursive subsystem $\mathcal{R}|_{\mathcal{A}_0}$.

Proposition 8.2. *Recursive subsystems are precisely strict subobjects represented by inclusions of descendant-closed atom sets.*

Definition 8.3 (Subsystem generated by a set). For $S \subseteq \mathcal{A}$, define

$$\langle S \rangle_{\text{rec}} = \bigcup_{\alpha \in S} \text{Desc}(\alpha).$$

This is the smallest recursive subsystem containing S .

Corollary 8.4. *Every finitely generated subsystem of an RRFS is finite if every atomic tree is finite, even when the ambient atom set is infinite.*

Definition 8.5 (Co-hereditary subset). A subset $U \subseteq \mathcal{A}$ is co-hereditary if

$$\beta \in U, C_{\beta, \alpha} > 0 \implies \alpha \in U.$$

Equivalently, it is upward-closed in the descendant order.

Thus hereditary and co-hereditary subsets play the roles of closed and open subspaces.

9 Quotients, congruences, and normality

A naive quotient that identifies arbitrary atom types need not preserve the predecessor operation.

Definition 9.1 (Recursive congruence). An equivalence relation $\sim$ on $\mathcal{A}$ is a recursive congruence if, whenever $\alpha \sim \alpha'$, then for every equivalence class K ,

$$\sum_{\beta \in K} B_{\beta, \alpha} = \sum_{\beta \in K} B_{\beta, \alpha'}. \quad (24)$$

Theorem 9.2 (Quotient system). A recursive congruence defines a quotient RRFs on $\mathcal{A}/\sim$ by

$$\bar{B}_{K,L} = \sum_{\beta \in K} B_{\beta, \alpha}, \quad \alpha \in L. \quad (25)$$

The definition is independent of the representative α , and the quotient map is a strict morphism.

This is analogous to an equitable partition of a directed multigraph and to a bisimulation-compatible quotient of a transition system.

9.1 Kernel congruences

Every strict morphism $f : \mathcal{R} \rightarrow \mathcal{S}$ defines

$$\alpha \sim_f \alpha' \iff f(\alpha) = f(\alpha').$$

Proposition 9.3. The kernel relation $\sim_f$ is a recursive congruence. The morphism factors as

$$\mathcal{R} \twoheadrightarrow \mathcal{R}/\sim_f \hookrightarrow \mathcal{S}$$

when the induced map from classes to the image atom set is injective.

9.2 Normal subsystems

The group-theoretic word “normal” has no unique direct analogue for atom subsets. Two useful notions should be distinguished.

Definition 9.4 (Categorically normal linear subsystem). In **LinRRFS**, a linear subobject $N \hookrightarrow R$ is *normal* if it is the kernel of a linear recursive morphism.

Because $HB_R = B_S H$, the kernel is B_R -invariant:

$$x \in \ker H \implies B_R x \in \ker H.$$

Thus normal linear subsystems are invariant coordinate submodules compatible with recursion.

Definition 9.5 (Saturated recursive subsystem). A descendant-closed atom subset $\mathcal{A}_0$ is saturated if every atom outside $\mathcal{A}_0$ has either no descendants in $\mathcal{A}_0$ or its entire predecessor multiset projects to zero in the chosen quotient. Equivalently, the collapse of $\mathcal{A}_0$ to a distinguished zero class defines a recursive congruence.

Remark 9.6. For most structural arguments, recursive congruences and kernel objects are cleaner than forcing a single group-like definition of normality.

10 Direct sums, products, and extensions

10.1 Direct sums and coproducts

For disjoint systems $\mathcal{R}_1, \mathcal{R}_2$, define

$$\mathcal{R}_1 \oplus \mathcal{R}_2$$

by disjoint union of atoms and block-diagonal B . In the strict category this serves as both product and coproduct under the expected componentwise maps; it is a biproduct-like construction at the coordinate level.

10.2 A pair product

One may form atoms $(\alpha, \gamma) \in \mathcal{A}_1 \times \mathcal{A}_2$. Two natural transition operators are:

$$B_{\boxtimes} = B_1 \otimes B_2,$$

or the asynchronous product

$$B_{\square} = B_1 \otimes I + I \otimes B_2. \tag{26}$$

For the asynchronous product, the rank

$$\rho(\alpha, \gamma) = \rho_1(\alpha) + \rho_2(\gamma)$$

strictly decreases along every edge.

Remark 10.1. The tensor product $B_1 \otimes B_2$ makes both components descend simultaneously and can terminate prematurely when one factor is terminal. The asynchronous product is usually the more robust categorical construction.

10.3 Extensions

A one-way coupled system

$$B = \begin{pmatrix} B_N & K \\ 0 & B_Q \end{pmatrix}$$

can be viewed as an extension of Q by N . There is a sequence

$$0 \longrightarrow N \longrightarrow R \longrightarrow Q \longrightarrow 0$$

at the level of coordinate modules, with the inclusion and projection intertwining transition operators.

Definition 10.2 (Split extension). The extension is split if there exists a recursive section $Q \rightarrow R$ of the projection. Block diagonal systems are split with $K = 0$.

The coupling matrix K measures the failure of the extension to decompose as a direct sum.

11 Connectedness, decomposition, and simplicity

Definition 11.1 (Connected RRFS). An RRFS is connected if the underlying undirected graph of its transition DAG is connected.

Theorem 11.2 (Canonical component decomposition). *Every finite RRFS decomposes uniquely, up to permutation, as a direct sum of connected RRFS corresponding to the connected components of its underlying undirected transition graph.*

Definition 11.3 (Indecomposable). An RRFS is indecomposable if it is not isomorphic to a nontrivial direct sum.

For finite systems, connectedness and indecomposability are equivalent.

11.1 Three notions of simplicity

The analogy with simple groups suggests several non-equivalent notions.

Definition 11.4 (Congruence-simple). An RRFS is congruence-simple if its only recursive congruences are equality and the indiscrete congruence, when the latter yields a ranked quotient.

Definition 11.5 (Subsystem-simple relative to terminals). Let $T \subseteq \mathcal{A}$ be the terminal atom set. The system is subsystem-simple relative to T if the only descendant-closed subsets containing T are T and $\mathcal{A}$.

Definition 11.6 (Linearly simple). A finite RRFS is linearly simple if the transition representation on $\mathbb{Q}^{\mathcal{A}}$ has no nonzero proper coordinate subspace invariant under B and compatible with the nonnegative cone.

Remark 11.7. Because every finite nilpotent operator has nontrivial invariant subspaces over a field, unrestricted module-theoretic simplicity is usually impossible. The cone-compatible or atom-compatible restrictions are essential.

Example 11.8. A two-level star with one nonterminal root type and a set of terminal types can be subsystem-simple relative to its terminal set, but it need not be congruence-simple if terminal types have identical structural roles.

12 Series and finite length

Definition 12.1 (Recursive filtration). A recursive filtration is a chain of descendant-closed subsystems

$$\emptyset = \mathcal{R}_0 \subseteq \mathcal{R}_1 \subseteq \cdots \subseteq \mathcal{R}_m = \mathcal{R}.$$

The rank filtration

$$\mathcal{A}_{\leq k} = \{\alpha : \rho(\alpha) \leq k\}$$

is canonical and descendant-closed.

Definition 12.2 (Composition series). A composition series is a finite filtration in which each quotient $\mathcal{R}_i/\mathcal{R}_{i-1}$ is simple in a specified chosen sense.

Remark 12.3. A Jordan–Hölder theorem is not automatic because the available notions of quotient and simplicity differ. Establishing appropriate hypotheses under which factor multisets are invariant is a natural research problem.

13 Complete fronts and polynomial functors

Let F be a finite rooted forest. A complete front is an antichain meeting every root-to-leaf path exactly once. Equivalently, it is a maximal antichain.

13.1 Univariate front polynomial

For a rooted tree T , define

$$C_T(x) = \sum_{A \in \text{CF}(T)} x^{\lambda(A)},$$

where $\lambda(A)$ counts selected terminal vertices. Then

$$C_T(x) = \begin{cases} x, & T \text{ is a single terminal vertex,} \\ 1 + \prod_{i=1}^k C_{T_i}(x), & T \text{ has child subtrees } T_i. \end{cases} \quad (27)$$

For forests,

$$C_{T_1 \sqcup \dots \sqcup T_r}(x) = \prod_{i=1}^r C_{T_i}(x).$$

Theorem 13.1 (General front bounds). *For a finite rooted forest F , let $R(F)$ be its number of components, $L(F)$ its number of leaves, $I(F)$ its number of non-leaf vertices, and $A(F) = \# \text{CF}(F)$. Then*

$$L(F) = R(F) + \sum_{v \text{ non-leaf}} (d^+(v) - 1), \quad (28)$$

$$R(F) \leq |A| \leq L(F) \quad (A \in \text{CF}(F)), \quad (29)$$

$$I(F) + 1 \leq A(F) \leq 2^{I(F)}. \quad (30)$$

13.2 Conserved weights

Definition 13.2. A weight $w : V(F) \rightarrow \mathcal{R}$ is front-conserved if

$$w(v) = \sum_{u \text{ child of } v} w(u)$$

for every non-leaf v .

Theorem 13.3 (Cut conservation). *For every complete front A ,*

$$\sum_{v \in A} w(v) = \sum_{r \text{ root}} w(r). \quad (31)$$

Taking $w(v)$ equal to the number of descendant leaves yields

$$\sum_{v \in A} L(v) = L(F).$$

Assigning transition probabilities yields a Kraft-type equality.

13.3 Probability and cumulants

Normalize

$$G_F(s) = \frac{C_F(s)}{C_F(1)}.$$

This is the probability generating function of the terminal count in a uniformly chosen complete front. For disjoint forests,

$$G_{F \sqcup F'}(s) = G_F(s)G_{F'}(s),$$

so the random counts add independently. Therefore every cumulant is additive:

$$\kappa_r(F \sqcup F') = \kappa_r(F) + \kappa_r(F'). \quad (32)$$

Furthermore,

$$\begin{aligned} \mathbb{E}[X] &= \frac{C'_F(1)}{C_F(1)}, \\ \text{Var}(X) &= \frac{C''_F(1)}{C_F(1)} + \frac{C'_F(1)}{C_F(1)} - \left(\frac{C'_F(1)}{C_F(1)} \right)^2. \end{aligned}$$

If $R(F) \leq X \leq L(F)$, then

$$\text{Var}(X) \leq \frac{(L(F) - R(F))^2}{4}.$$

14 Hadamard closure and overlap semirings

Lemma 14.1 (Integral cone inequality). *Let $a_j \in \mathbb{N}_0$ and suppose*

$$X \geq \sum_j a_j x_j, \quad Y \geq \sum_j a_j y_j$$

for nonnegative x_j, y_j . Then

$$XY \geq \sum_j a_j x_j y_j. \quad (33)$$

Proof. Put $U = \sum_j a_j x_j$ and $V = \sum_j a_j y_j$. Then

$$UV - \sum_j a_j x_j y_j = \sum_{j \neq k} a_j a_k x_j y_k + \sum_j a_j (a_j - 1) x_j y_j \geq 0.$$

□

Theorem 14.2 (Hadamard closure). *For every RRFs with nonnegative integer transition matrix,*

$$x, y \in \mathcal{C}_{\mathcal{R}} \implies x \circ y \in \mathcal{C}_{\mathcal{R}},$$

where $(x \circ y)_{\alpha} = x_{\alpha} y_{\alpha}$.

Proof. For each coordinate β , the cone inequalities give

$$x_{\beta} \geq \sum_{\alpha} B_{\beta, \alpha} x_{\alpha}, \quad y_{\beta} \geq \sum_{\alpha} B_{\beta, \alpha} y_{\alpha}.$$

Apply the lemma with $a_{\alpha} = B_{\beta, \alpha}$.

□

Thus the object monoid acquires an overlap product $\star_{\mathcal{R}}$ defined by

$$M(a \star_{\mathcal{R}} b) = M(a) \circ M(b).$$

Ordinary multiplication becomes coordinate addition, and $\star_{\mathcal{R}}$ becomes coordinatewise multiplication. Hence

$$a \star_{\mathcal{R}} (bc) = (a \star_{\mathcal{R}} b)(a \star_{\mathcal{R}} c).$$

The support identity is

$$\text{supp } M(a \star_{\mathcal{R}} b) = \text{supp } M(a) \cap \text{supp } M(b).$$

15 Arithmetic and polynomial morphisms

15.1 The prime-index isomorphism

Let $\mathcal{R}_{\mathbb{N}}$ be the numerical Pratt system and let $\mathcal{R}_f$ be the polynomial subsystem with atoms $\{f_p : p \in \mathbb{P}\}$. Define

$$F(p) = f_p.$$

Then

$$D(F(p)) = Df_p = f_{p-1} = \prod_{q|p-1} f_q^{v_q(p-1)} = \widehat{F}(Dp).$$

Therefore F is a strict isomorphism. Its coordinate consequence is

$$M_{f_q}(f_n) = m_q(n). \tag{34}$$

Evaluation at $x = 2$ satisfies

$$\text{ev}_2(f_p) = p, \quad \text{ev}_2(Df_p) = p - 1,$$

and is the inverse strict morphism on this subsystem.

15.2 Embedding into the general polynomial system

The inclusion

$$\{f_p\} \hookrightarrow \mathcal{I}_{\mathbb{Q}}$$

is a recursive subobject. Thus the chain

$$\mathcal{R}_{\mathbb{N}} \cong \mathcal{R}_f \hookrightarrow \mathcal{R}_{\text{poly}, \mathbb{Q}}$$

exhibits the numerical system as a concrete recursively closed polynomial subsystem.

16 Software versioning and dependency provenance

Let atoms be exact package-version identifiers, not merely package names. This preserves deterministic predecessor data. For each package version P , define

$$D(P) = \prod_Q Q^{b_{Q,P}}.$$

16.1 Root reconstruction and consistency

Given a total unfolded dependency count x , the root declaration vector is

$$u = (I - B)x.$$

A necessary and sufficient condition for x to be a valid occurrence vector is

$$x \geq 0, \quad (I - B)x \geq 0. \quad (35)$$

Negative entries certify inconsistency between the observed dependency inventory and the assumed architecture.

16.2 Version abstraction as quotient

Mapping exact versions to package families defines a quotient only if versions identified in one class have equal aggregated dependency multiplicities into every target class. This is exactly the recursive congruence condition. If the condition fails, the abstraction is lossy in a way that does not define an RRFS quotient.

16.3 Release sequences and filtrations

A sequence of software architectures

$$\mathcal{R}_1 \hookrightarrow \mathcal{R}_2 \hookrightarrow \dots$$

can model successive releases when every old package-version type retains its predecessor data and new types are added above or alongside existing ones. The direct limit is an infinite locally ranked RRFS if every atom has finite ancestry.

16.4 Shared exposure and similarity

For products A, B , define

$$\begin{aligned} K(A, B) &= \langle M(A), M(B) \rangle, \\ O(A, B) &= M(A) \circ M(B). \end{aligned}$$

The first is a positive-semidefinite dependency kernel. The second is a valid recursive occurrence profile whose support is exactly the set of shared transitive dependency types. Criticality weights $w_Q \geq 0$ give

$$K_w(A, B) = \sum_Q w_Q M_Q(A) M_Q(B).$$

16.5 Control fronts

Complete fronts in an unfolded dependency tree are maximal antichains that meet every root-to-terminal path exactly once. They model checkpoint layers. With checkpoint costs $c(v)$, the minimum complete-front cost satisfies the dynamic recursion

$$V(T) = \min \left\{ c(r), \sum_i V(T_i) \right\}. \quad (36)$$

The full front polynomial records not only one optimum but the entire distribution of admissible checkpoint layers.

16.6 Limitations

Real package managers include optional dependencies, platform conditions, version ranges, deduplication, conflicts, and cycles. The basic RRFS applies exactly after choosing a deterministic resolved dependency DAG and unfolding occurrences. Generalizations may replace fixed predecessor monomials by sets, probabilities, semiring weights, or time-dependent operators.

17 Further structural analogies

Classical language	RRFS analogue	Caution
Subspace	Descendant-closed subsystem	Exact via Alexandrov topology
Open subset	Co-hereditary atom set	Upward-closed in descendant order
Continuous map	Strict recursive morphism	Order-preserving automatically
Subgroup/submodule	Recursive subobject / invariant coordinate submodule	Two levels should be distinguished
Normal subgroup	Kernel object or recursive congruence	No unique atom-level analogue
Quotient group	Quotient by recursive congruence	Requires equitable predecessor multiplicities
Direct product	Disjoint direct sum or pair product	Several inequivalent products exist
Group extension	Block-triangular coupled system	Coupling matrix is extension data
Simple object	Congruence-simple or subsystem-simple	Notions are non-equivalent
Connected space	Connected transition graph	Equivalent to indecomposable for finite systems
Composition series	Filtration by recursive subsystems	Jordan–Hölder remains open
Metric space	Hilbert metric on total coordinates	Multiplication acts by translations
Inverse/direct limit	Compatible chains of systems	Local finiteness must be preserved

18 Open problems

Question 18.1 (Categorical limits). Which small limits and colimits exist in **RRFS** and **LinRRFS**? Under what hypotheses do products, equalizers, coequalizers, and pushouts preserve ranked local finiteness?

Question 18.2 (Jordan–Hölder theory). Is there a useful class of finite RRFS and a preferred simplicity notion for which composition factors are unique up to permutation and isomorphism?

Question 18.3 (Classification of simple systems). Classify congruence-simple finite acyclic multiplicity systems. How does the answer relate to equitable partitions and automorphisms of the transition DAG?

Question 18.4 (Automorphism groups). Determine

$$\text{Aut}_{\mathbf{RRFS}}(\mathcal{R}) = \{f : \mathcal{A} \rightarrow \mathcal{A} : f \text{ bijective and } f_*B = Bf_*\}.$$

For finite systems this is the automorphism group of the weighted transition DAG.

Question 18.5 (Exactness). Develop an exact-category or proto-abelian structure on a suitable linear enlargement of $\mathbf{LinRRFS}$. Which kernel-cokernel pairs correspond to genuine recursive quotients?

Question 18.6 (Topological completion). For infinite systems, what completions of $\ell^2(\mathcal{A})$ or weighted sequence spaces make B , C , and morphisms bounded? Can direct-limit RRFS be reconstructed from finite rank truncations?

Question 18.7 (Polynomial realization). Which finite RRFS can be realized by irreducible polynomials and a predecessor operator $D\varphi = \varphi - \varphi(0)$ over some field? Which can be realized as a recursively closed subsystem of prime-index polynomials or their analogues?

Question 18.8 (Software experiments). Do RRFS coordinate kernels and boundary consistency tests improve vulnerability clustering, release comparison, and dependency provenance relative to ordinary package-set or graph-embedding methods?

19 Conclusion

Ranked recursive factorization systems isolate a common mechanism behind numerical Pratt trees, prime-index polynomials, general polynomial Pratt forests, finite-field truncations, finite DAGs, and deterministic dependency architectures. The central equations

$$M = v + BM, \quad v = (I - B)M, \quad M = (I - B)^{-1}v$$

are simultaneously combinatorial counting identities, coordinate transforms, and natural transformations. The telescoping formula

$$a = \prod_{\alpha} \left(\alpha D(\alpha)^{-1} \right)^{M_{\alpha}(a)}$$

shows that recursive coordinates retain enough information to reconstruct the original factorization object. Complete fronts add generating functions, probability laws, inequalities, and conservation principles; integral transition multiplicities add Hadamard closure and overlap arithmetic.

The categorical viewpoint clarifies which constructions are intrinsic. Descendant-closed subsets are subobjects and closed subspaces of the Alexandrov topology; recursive congruences produce quotients; kernels provide a precise linear notion of normality; disjoint unions give canonical decompositions; block-triangular systems are extensions; and strict morphisms transport the complete recursive structure. In particular, the isomorphism $p \mapsto f_p$ is no longer merely an analogy between integers and polynomials: it is a morphism preserving predecessor factorizations, trees, coordinates, boundary, incidence, and reconstruction.

The next stage is a structural theory of finite RRFS—classification of simple and indecomposable systems, automorphism groups, quotient lattices, composition series, exactness, and realization—together with empirical testing on software dependency and other engineered hierarchies.

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